



ABES Engineering College, Ghaziabad



I N D E X

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Roll No. : 2400320100636

S. No.	Name of the Experiment	Date of Practical		Page No. From To	Assessment by faculty member					
		Scheduled	Actual		Pre-Lab Writing Work* MM: 05	Implementation/ Active Participation MM: 10	Graph, Results/ Output, Calc. MM: 05	In time submission of Lab Report/ Viva Voce MM: 05	Total MM: 25	Signature With Date
	Study of various types of active & passive components	07/07/25	07/07/25		5	9	4	4	22	Ma 21/4
	Study of lab equipment & components	07/07/25	07/07/25		5	10	5	4	24	Ma 21/4
	Verification of truth table of various logic gates	21/07/25	21/07/25		5	10	5	4	24	Ma 21/4
	Implementation of given boolean function using SOP & POS	21/07/25	21/07/25		5	10	4	4	23	Ma 28/4
	To study the V-I characteristics of P-N junction diode	28/07/25	28/07/25		5	10	5	5	25	Ma 12/5
	To study the V-I characteristics of Zener diode	28/07/25	28/07/25		5	10	5	5	25	Ma 12/5
	Application of P-N junction diode	12/5/25	12/5/25		5	10	5	5	25	Ma 12/5
	To study the operational amplifier as adder & subtractor	12/5/25	12/5/25		5	10	5	4	24	Ma 19/5
	To study & implement Zener diode as voltage regulator	19/5/25	19/5/25		5	10	5	4	24	Ma 26/5
	To study the working of BJT as switch	19/5/25	19/5/25		5	10	5	4	24	Ma 21/6
	Soldering shop	26/5/25	26/5/25		5	9	4	4	22	Ma 21/6

's Remarks (if any):

Good work. Keep it up

Average Marks:

23.81

n. of Faculty member(s) with date

Mr. Manish Zadoo Mas 21/6/25

Name & Sign. of Lab In charge

HOD

Lab writing work should include problem statement, objective, Algorithm (if applicable) and Methodology.
members will check pre Lab writing work & Lab work readings/output in each class and sign with date.
pen with blue color Ink only.

EXPERIMENT NO. 1

(This page needs to be filled by pencil only)

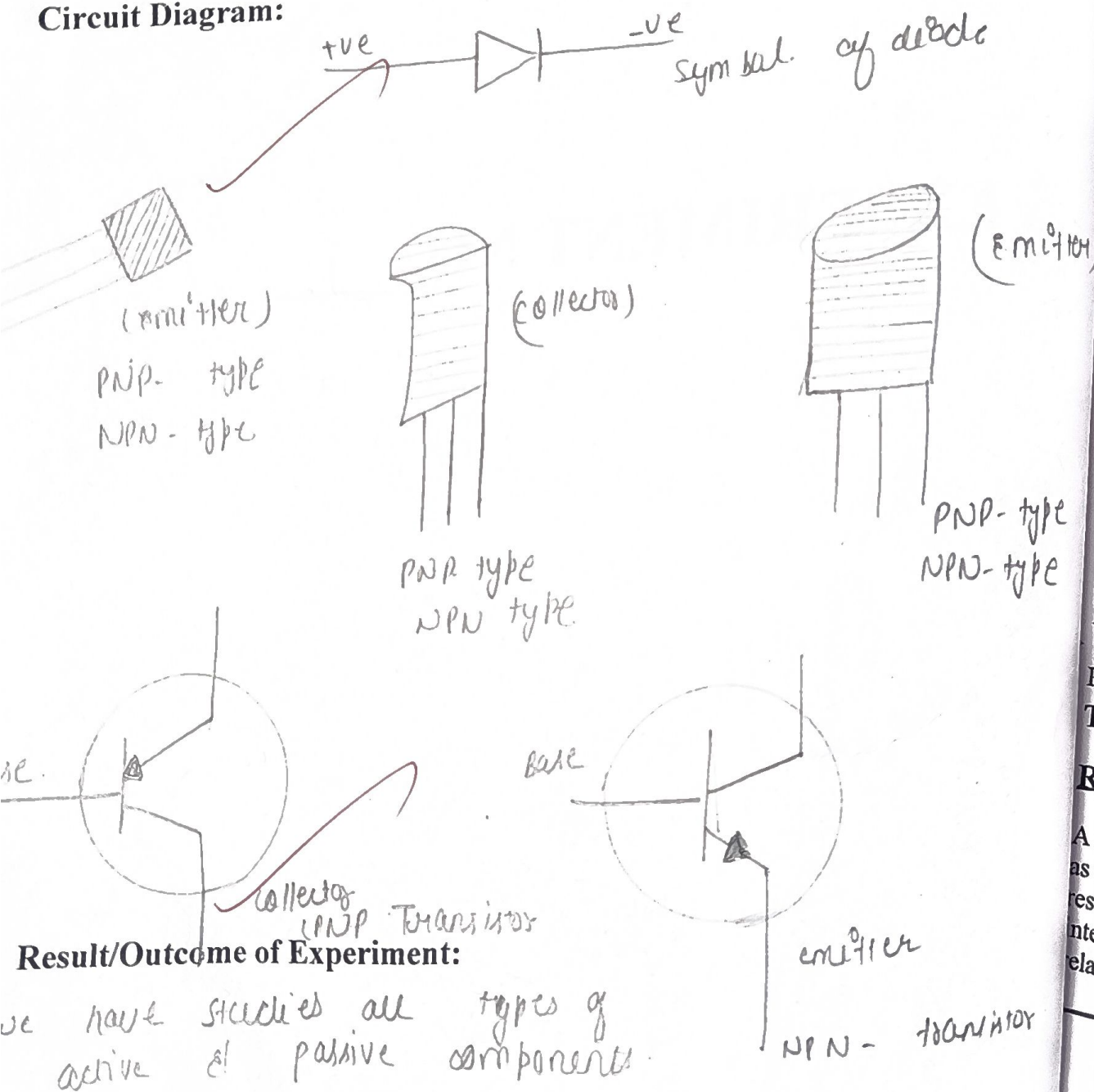
Aim of the Experiment:

Study of various types of active and passive components based on readings

Apparatus used:

Active and passive components.

Circuit Diagram:



Result/Outcome of Experiment:

We have studied all types of active & passive components.

AIM: Study of various types of active and passive components based on readings

APPARATUS: Active and passive components

THEORY:

ACTIVE COMPONENTS

(a) Transistor

A transistor is a semiconductor device that can amplify or switch electronic signals and electrical power. It is composed of semiconductor material with at least two terminals for connection to an external circuit. For example, a transistor with a base, emitter, and collector terminal. The operation of a transistor is based on the movement of electrons and holes within the semiconductor material.

PASSIVE COMPONENTS

Resistors of various types

THEORY:-

RESISTANCE:-

A resistor is a passive component in an electrical circuit that opposes the flow of current through it. It is represented by a zigzag line in a circuit diagram. The resistance of a resistor is measured in ohms (Ω). The relationship between the voltage across a resistor and the current through it is given by Ohm's Law: $V = IR$, where V is the voltage, I is the current, and R is the resistance.

separation in a parallel-plate capacitor causes an internal electric field. A dielectric (orange) reduces the field and increases capacitance. A capacitor consists of two conductors separated by a non-conductive region. The non-conductive region is called the dielectric. In simpler terms, the dielectric is just an electrical insulator. Examples of dielectric media are glass, air, paper, vacuum, and even a semiconductor depletion region chemically identical to the conductors. A capacitor is assumed to be self-contained and isolated, with no net electric charge and no influence from any external electric field. The conductors thus hold equal and opposite charges on their facing surfaces, and the dielectric develops an electric field. In SI units, a capacitance of one farad means that one coulomb of charge on each conductor causes a voltage of one volt across the device. The capacitor is a reasonably general model for electric fields within electric circuits. An ideal capacitor is wholly characterized by a constant capacitance C , defined as the ratio of charge $\pm Q$ on each conductor to the voltage V between them: $C = Q/V$. Sometimes charge build-up affects the capacitor mechanically, causing its capacitance to vary. In this case, capacitance is defined in terms of incremental changes:

Capacitance Measurement

A capacitor is a device that stores energy. It is widely used to give a boost of energy at start up when power is applied to lighting and motor systems. To test a capacitor

- Set the function switch to the desired F(capacitance) range.
- Insert the capacitor leads directly in to Cx socket.
- Read the capacitance directly from display.

OBSEVATIONS:

S.N	Component	Value using color code (X _C)	Value using DMM (X _M)	Error X _C - X _M
1	Resistor 1			
2	Resistor 2	1000 Ω	990 Ω	10 Ω
3	Resistor 3	100 Ω	99.5 Ω	0.5 Ω
4	Capacitor 1	10.00 μF	9.92 μF	0.08 μF
5	Capacitor 2	470 PF	463 PF	7 PF
6	Inductor	100 μH	92.5 μH	7.5 μH
		2.2 μH	2.14 μH	0.06 μH

RESULT:

The value of the components are checked & verified.

PRECAUTIONS:

1. All connections should be tight and right.
2. Components should be handled carefully.

Questions:

Q.1 How do you identify active and passive components?

ANS: An active and passive components in an Experiment can be identified as if there is a decline in power the component is passive and if signal is amplified it is an active component.

Q.2 How many types of passive components are there?

ANS: Resistors of various values, inductors and various types of capacitors.

Q.3 Is led active or passive?

ANS: LED is considered as active components because they require a current to flow through them in order to produce light.

Q.4 What are the examples of active components?

ANS: Examples of active components are:
Diode, Integrated circuits, Transistors.

Q.5 What is active and passive elements give examples?

ANS: Active elements are the elements which are capable to delivering energy to the circuit.

Eg → Diode, Transistors.

Passive Components are capable of storing energy in the form of voltage or current in the circuit.

Eg → Resistors, capacitors and inductors.

Marks Obtained	22
Faculty Signature	Maw

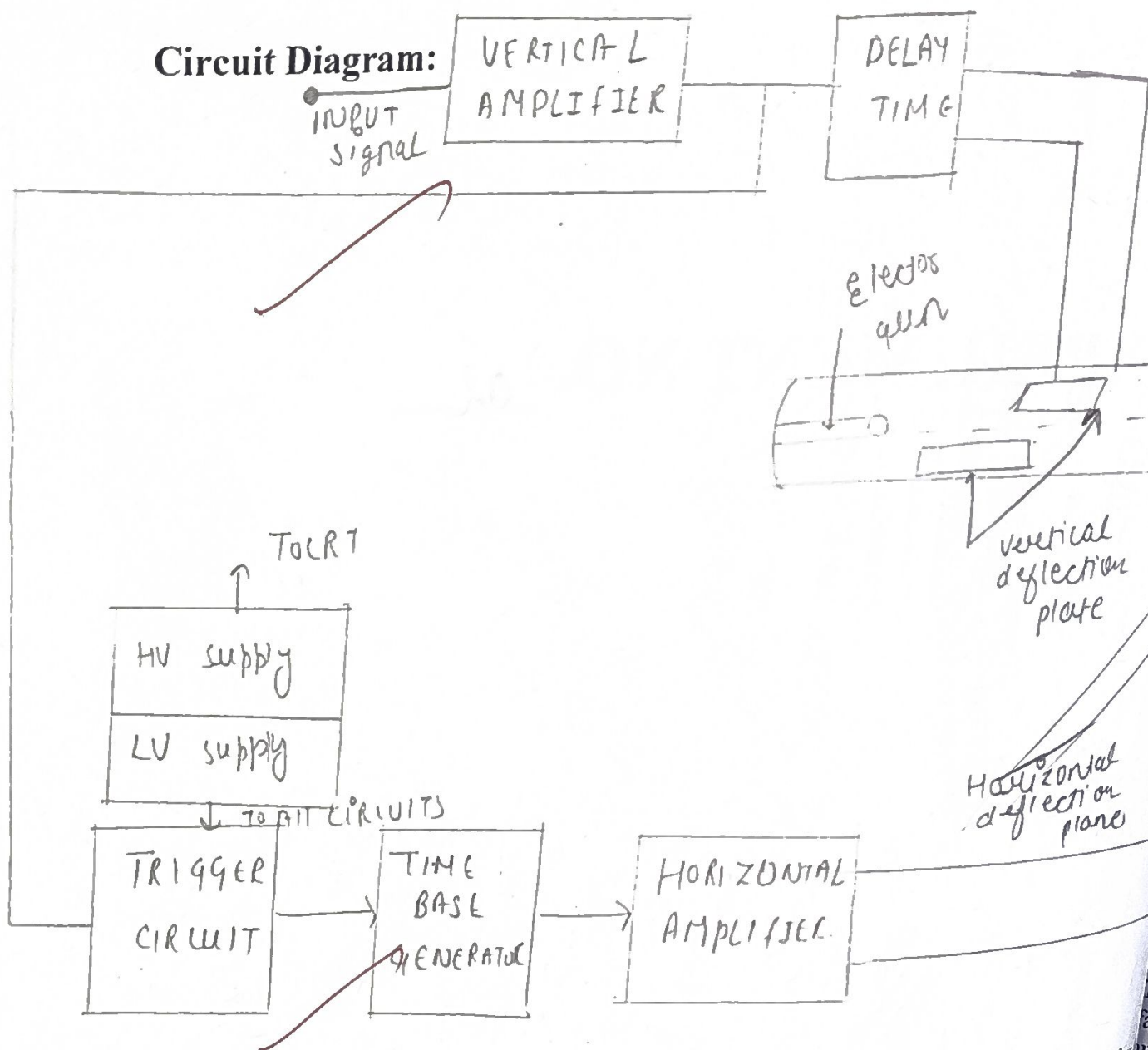
EXPERIMENT NO. 2

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Aim of the Experiment: Study of lab equipments and components
CRO, multimeter and function generator, power supply
Active, passive components and bread board

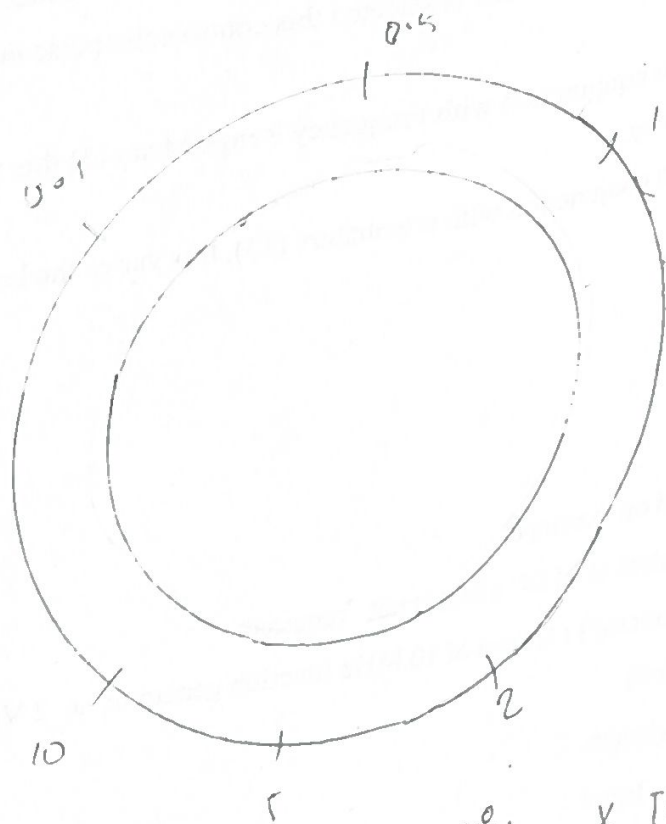
Apparatus used: CRO, multimeter and function generator,
supply - active, passive components and the bread

Circuit Diagram:



Result/Outcome of Experiment:

we have studied the
experimental and components.
CRO, multimeter, function generator, power



Time/div

T = number of horizontal div. x Time/div

$$= 4 \times 0.5 \text{ ms/div}$$

$$= \boxed{T = 2 \text{ ms}}$$

$$f = \frac{1}{T} = \frac{1}{2 \text{ ms}} = 500 \text{ Hz}$$

unc
The n
wavef
Signal
Conne
attenuat
help of
maximum
to use the
overshoot

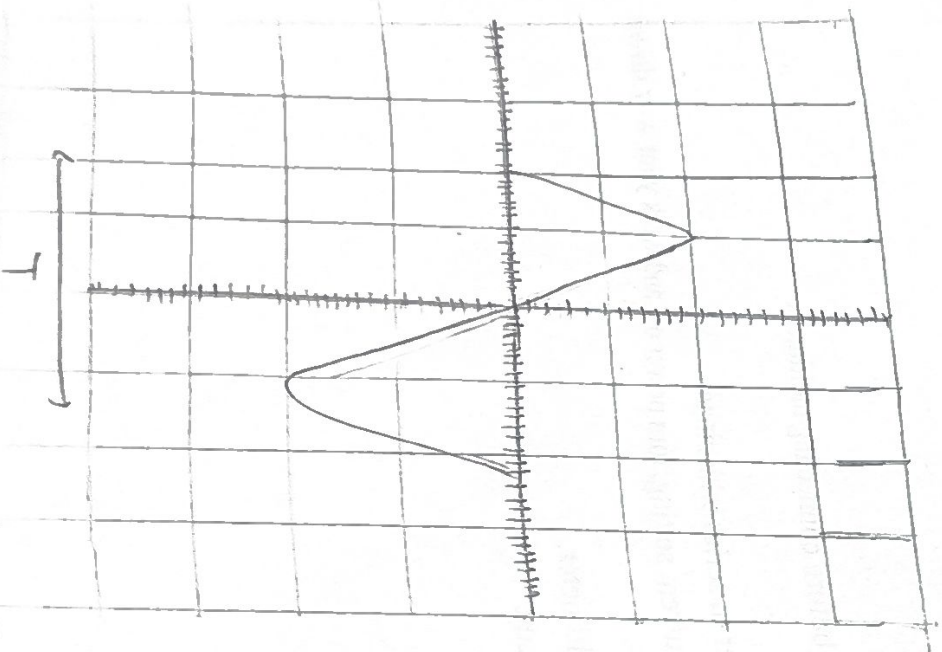
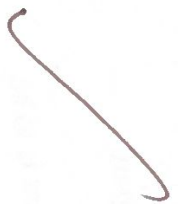
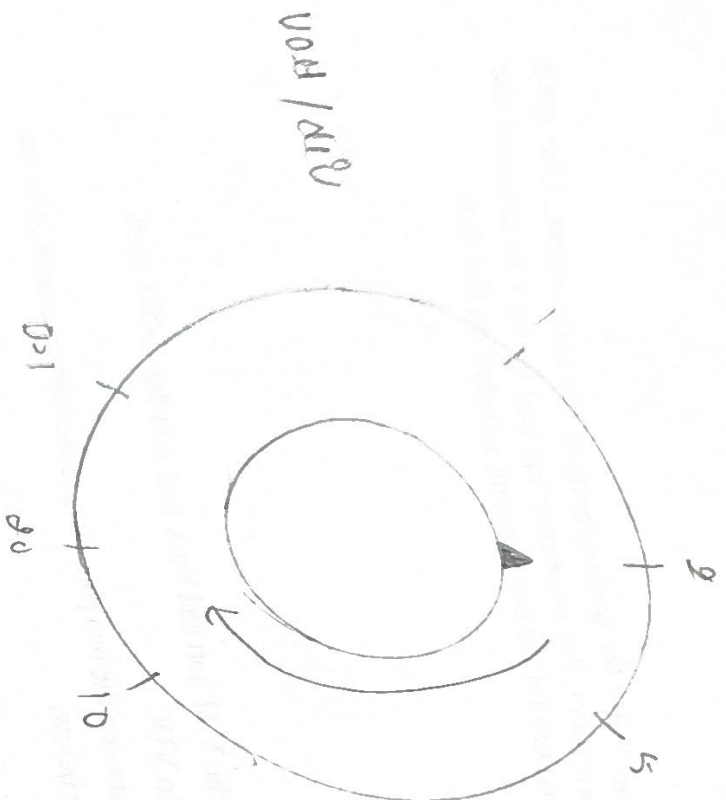
FM Input M
The Caddo
modulation l
modulation s
level not exce
The user can s
MHz generator
is available at
frequency modu
Keep maximum

DIGITAL MULT

THEORY: A me
voltmeter measure
ohmmeter measure
some additional ones
numbers, usually on a

Setting the Range

The dial also plays and
range of measurement.
the decimal point as it a
determines how refined,



RECAUT

- Do
- open
- Never

RESU

100: no of vertical div. x
4 div x 2 volt/div
8 volts

$V_p = 4 \text{ volts}$

OBSERVATION: S. No.	Name of the component	Measured value (using DMM)
1.	Diode	0.7V
2.	Transistor	—
3.	Transistor	—
4.	Transistor h_{fe}	300
5.	AC Voltage	220V
6.	DC Voltage	5V
7.	Current	1mA

RESULT: We have successfully studied various lab equipments and components like CRO, multimeter and junction generator, power supply - Active, passive components and bread board.

CAUTIONS:

Do not use DMM if its or test leads look damaged, or if you suspect that the meter is not operating properly.
Never ground yourself when taking electrical measurements.

- Turn off the power to the circuit under test before cutting, unsoldering or breaking the circuit. Small amount of current can be dangerous.
- When using the probes, keep your fingers behind guards on the probes.
- Measuring voltage, which exceeds the limits of the multimeter, may damage the meter and expose the operator to a shock hazard. Always recognize the meter voltage limits as stated on the meter.

Questions:

Q.1 What are the applications of C.R.O.

ANS: The CRO is used to provide accurate time and amplitude measurement of voltage signals over a wide range of frequencies.

Q.2 What are the applications of Digital Multimeter.

ANS: DMM is used as an ammeter, voltmeter and an ohmmeter.

Q.3 What are the applications of Function Generator.

ANS: It is used in generating different types of electrical equipment waveforms over a wide range of frequencies.

Q.4 What are the applications of Power supply.

ANS: Power supply is used to supply power in the circuit in the form of voltage.

Q.5 What is the difference between active and passive components.

- Active component deliver power in the circuit.
- Passive component can receive energy & store it in the form of magnetic fields.

Marks Obtained	24
Faculty Signature	Maw

EXPERIMENT NO. 3

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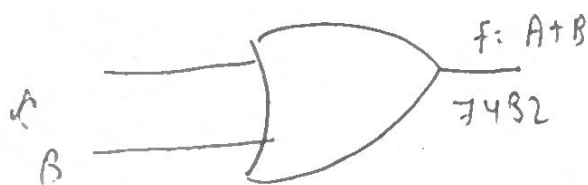
Aim of the Experiment:
Nomenclature data of

Introduction to digital electronics
digital IC's specifications, specifications
logic gates using TTL IC's

Apparatus used: IC's and bread board

Circuit Diagram:

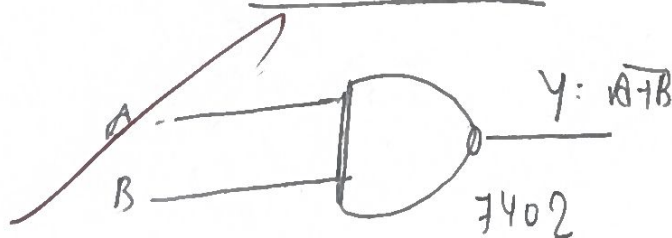
OR gate



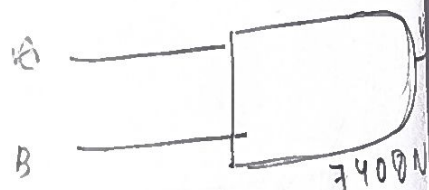
NOR gate



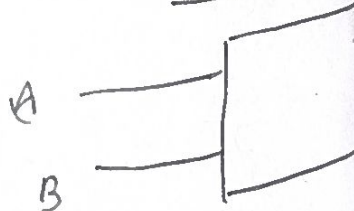
EX-NOR gate



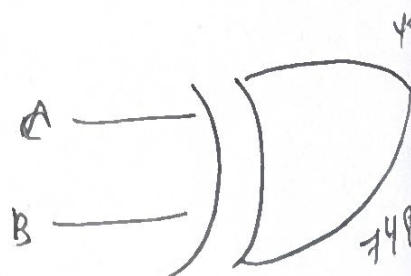
AND GATE



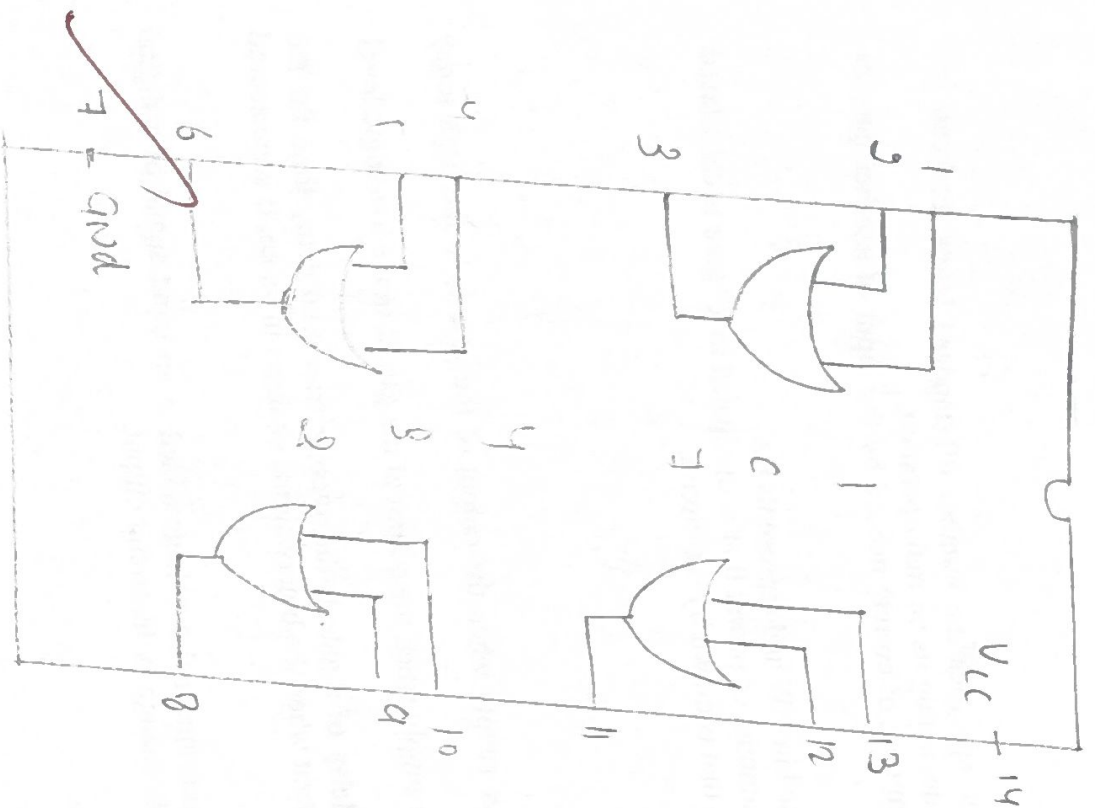
NAND gate



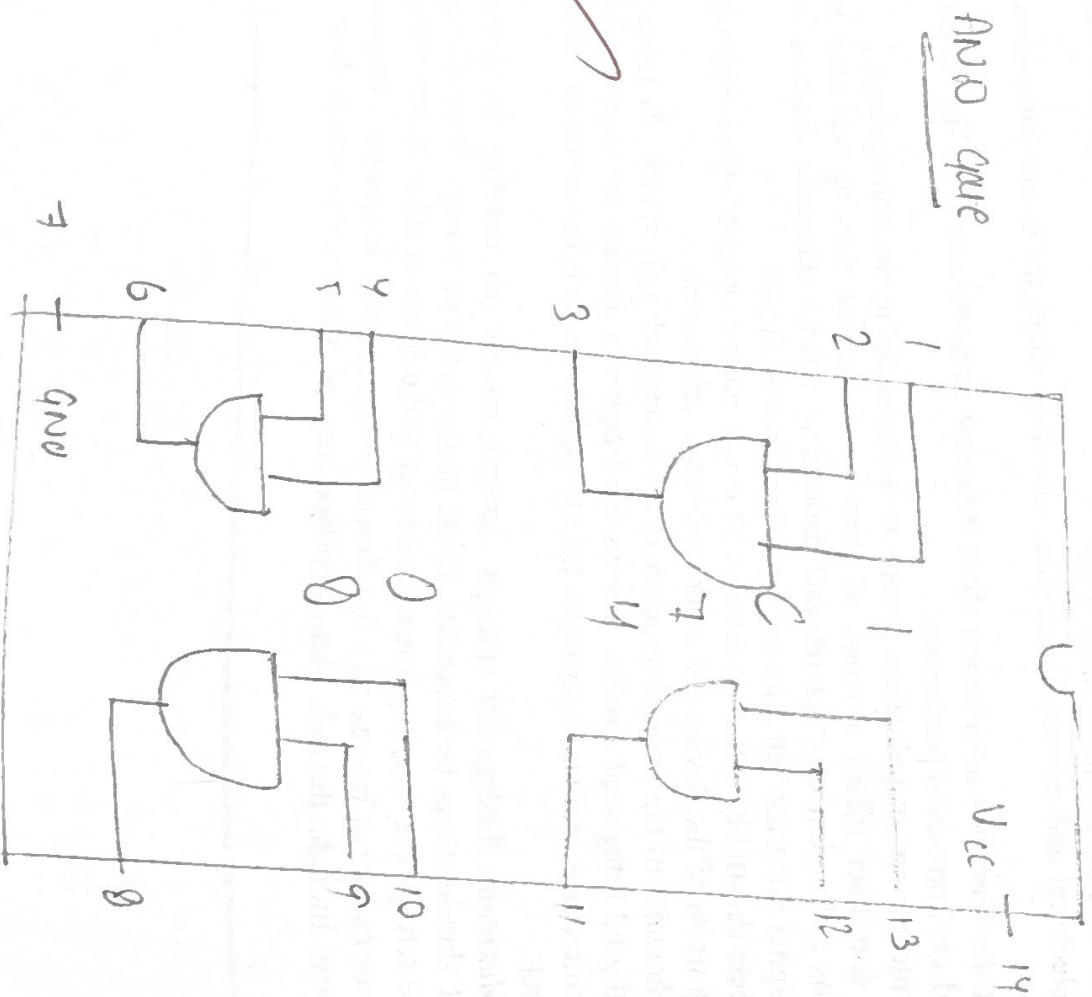
EXOR



Result/Outcome of Experiment: we have studied and the logic gates using truth table.



OR gate



AND gate

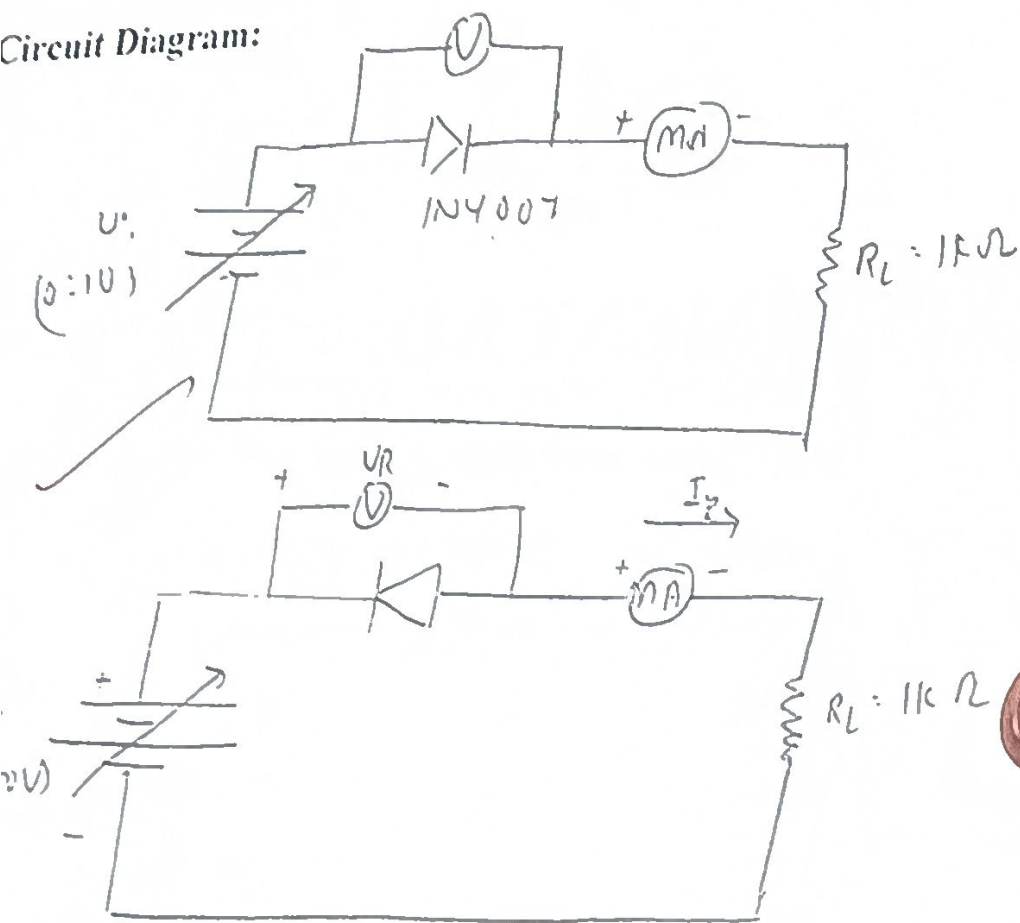
EXPERIMENT NO. 5

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Aim of the Experiment: *P-N junction diode: characteristics of P-N junction static & dynamic resistance measurement from graph*

Apparatus used: *Solder board, multimeter, variable DC power supply, resistance, diode & connecting wire*

Circuit Diagram:



Outcome of Experiment: *Static & dynamic resistances have been measured*

AIM: P-N Junction Characteristics Static and Dynamic

APPARATUS: 1kΩ, Diode (IN4007) THEORY:

When a p-type semiconductor will combine and result in positive & negative ions.

Forward Characteristics

The circuit arrangement of a diode is connected to a circuit so as to limit the current it may get permanently damaged. A milliammeter measures current connected to the anode of the diode in small steps about 0.1 V. The voltage across the diode is plotted with voltage across the diode to get a curve called the forward characteristic curve. The potential is reduced.

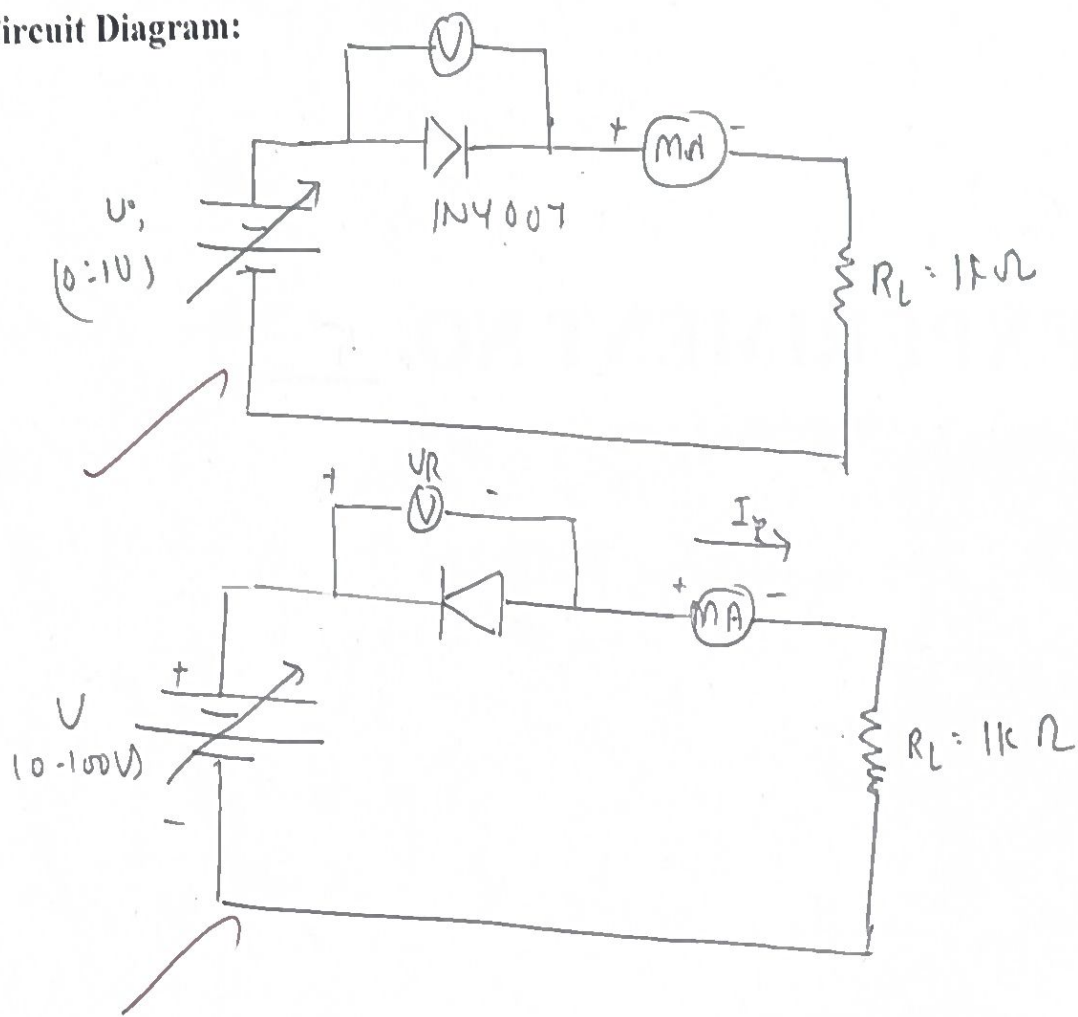
Static Resistance

EXPERIMENT NO. 5 (This page needs to be filled by pencil only)

Aim of the Experiment: P-N junction diode:
characteristics of P-N junction
static & dynamic resistance measurement from graph

Apparatus used: Bread board, multimeter, variable DC power supply, resistance, diode & connecting wire

Circuit Diagram:



Result/Outcome of Experiment: Static & dynamic resistances have been measured

AIM: P-N Junction Diode Characteristics

APPARATUS: (1kΩ), Diode, Breadboard, Multimeter, Variable DC power supply, Resistance, Diode & connecting wire

THEORY:
When a p-type semiconductor and an n-type semiconductor are joined together, a p-n junction is formed. The p-type semiconductor has an excess of holes (positive charge carriers) and the n-type semiconductor has an excess of electrons (negative charge carriers). When a voltage is applied across the junction, the positive charges move towards the negative terminal and the negative charges move towards the positive terminal, creating an electric current.

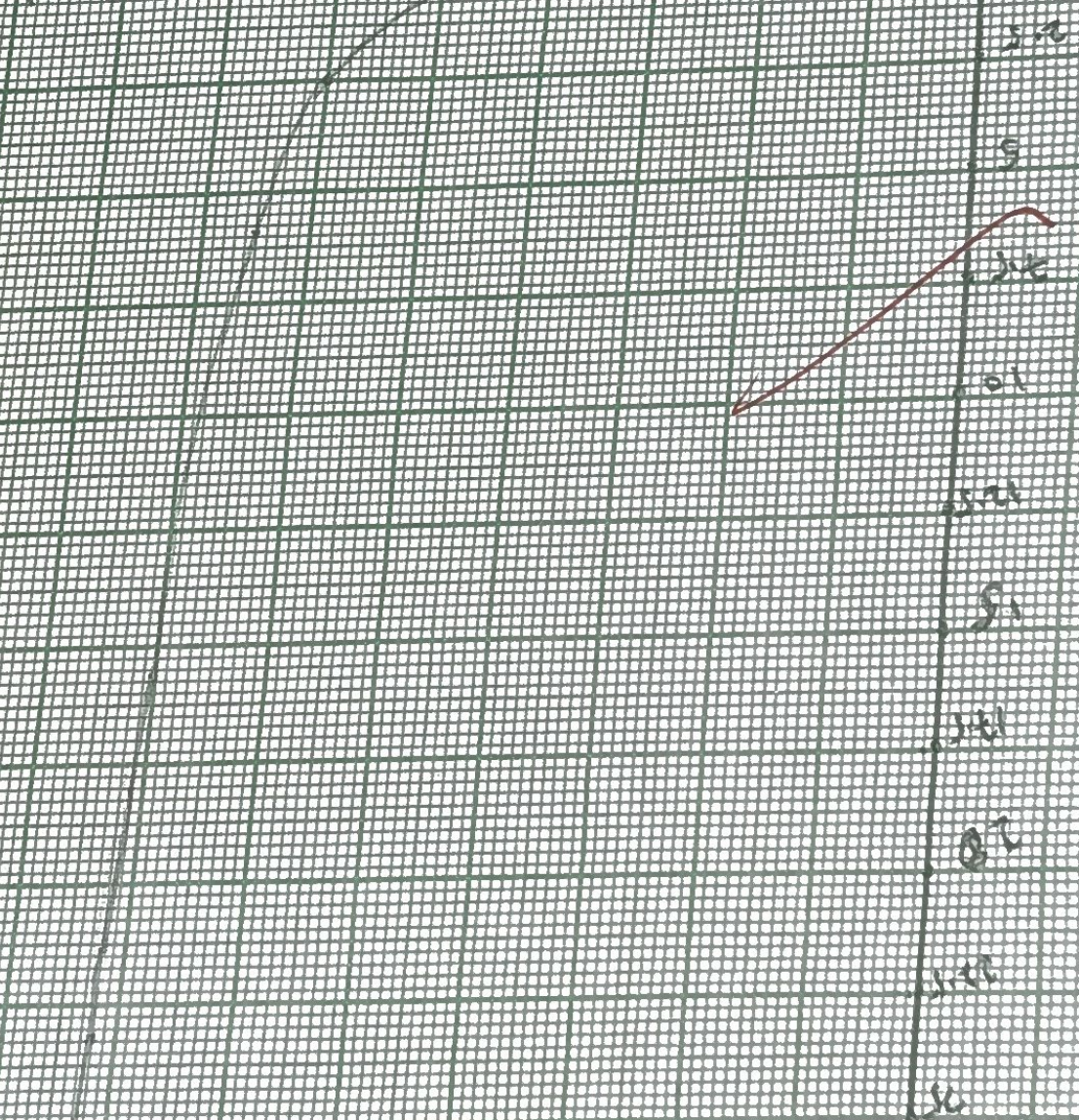
Forward Bias:

The circuit diagram for forward bias is shown below. The positive terminal of the DC power supply is connected to the p-type semiconductor (anode) and the negative terminal is connected to the n-type semiconductor (cathode). The diode is connected in series with a load resistor R_L and a milliammeter (mA). A voltmeter (V) is connected in parallel across the diode. The current flowing through the circuit is labeled I_F . The voltage across the diode is labeled V_F . The resistance of the load resistor is labeled $R_L = 1k\Omega$.

Static Resistance:

← $f(\omega)$

2.0 1.5 1.0 0.5 0 0.5 1.0 1.5 2.0



$f(\omega)$

→

0.5 1.0 1.5 2.0

0.5 1.0 1.5 2.0

0.5 1.0 1.5 2.0

0.5 1.0 1.5 2.0

0.5 1.0 1.5 2.0

Unit = 2.5 mm

y-axis

Unit = 0.1 V

x-axis

Scale

S.No.	V_D	Forward Bias		Reverse Bias	
		I_D (mA)	V_R (Volt)	I_R (mA)	
1	0.1	0	1	0	
2	0.2	0	5	0	
3	0.3	0	10	0	
4	0.45	0.1	15	0	
5	0.5	0.2	20	0	
6	0.6	2.5	25	0	
7	0.65	2.9	30	0	
8	0.7	5.4			
9	0.72	15.2			
10	0.72	22.8			
11	0.73	31.2			
		34.0			

CALCULATION: - Comparing practical and theoretical values.

RESULT: Static and dynamic resistances have been measured

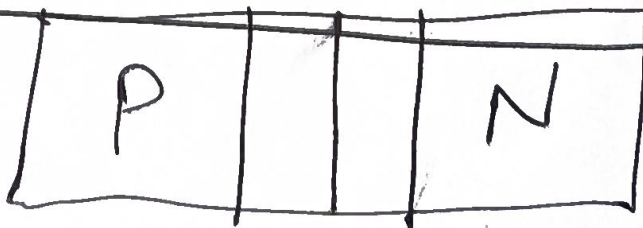
PRECAUTIONS:

- 1) All connections should be tight.
- 2) All steps should be followed carefully.
- 3) Readings should be taken carefully.
- 4) Do not touch live materials.
- 5) Assume a scale for V-I characteristic in forward & reverse biasing.
- 6) Never increase RB voltage up to breakdown voltage for an ordinary diode.

Questions:

Q.1 How many junctions does a diode have?

ANS: A diode consists of only one junction



Q.2 What are the major applications of pn junction?

ANS: It can be used as a photo diode, solar cell, LED lighting applications

Q.3 What devices use pn junctions?

ANS: They are used in elementary building blocks of semiconductor electronics devices such as diodes, transistors, solar cell

Q.4 What are the possible causes that may destroy a pn junction?

ANS: Increasing the voltage beyond 0.7V may result high enough value to destroy the diode.

Q.5 Why PN junction is called a diode?

ANS: P-N junction is called a diode because they allow the flow of current in one direction & also because there are two distinct electrode cathode anode.

Marks Obtained	25
Faculty Signature	Mao

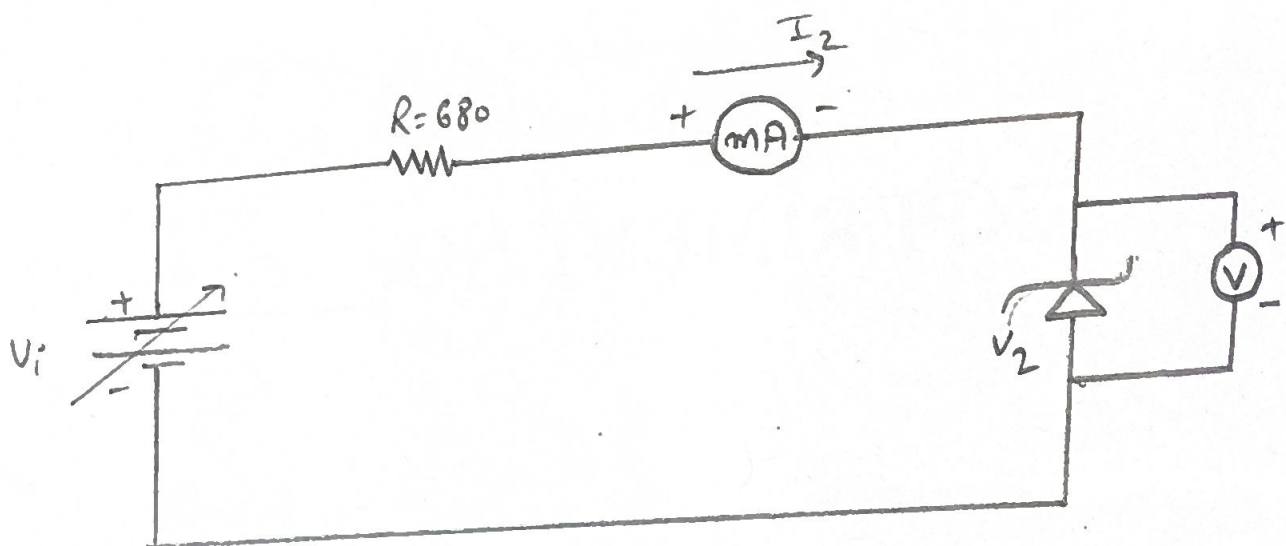
EXPERIMENT NO. 6

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Aim of the Experiment: Characteristics of Zener diode -
VI characteristics of Zener diode, graphical measurement of forward and reverse resistance.

Apparatus used:
Bread board, multimeter, regulated DC power supply, resistance
Zener diode, connecting wires.

Circuit Diagram:



Result/Outcome of Experiment: Forward and reverse current have been measured.

Air
V
a

Ap
B
C

Ci

RA

X axis: $I_{mA} = IV$
Y axis: $1 \text{ unit} = 2 \mu A$

Forward current

$V_i = 0.0V \rightarrow$ zero voltage

Reverse voltage

PRECA

1. A
2. M
3. Ta

Question
Q.1 Wha
ANS:

flow

OBSERVATIONS: V-I characteristics of Zener diode

S. No.	Reverse Voltage (Volt)	Reverse Current (Amp)	Forward Voltage (Volt)	Forward Current (Amp)
1	0	0	0.7	0.7
2	-2	0	0.72	1.8
3	-4.75	0	0.73	3.2
4	-5.8	-1.0	0.75	5.2
5	-5.87	-1.2	0.75	6.2
6	-5.9	-3.9	0.765	9.11
7	-5.94	-6.8	0.77	11.07
8	-5.97	-9.8	0.78	15.08
9	-6.01	-13.6	0.785	19.04
10	-6.06	-19.6	0.786	23.06

Forward Resistance: $R_F = \frac{V_F}{I_F}$

Reverse Resistance $R_r = \frac{V_r}{I_r}$

RESULT: we have obtained V-I characteristics of Zener diode as in graph with

$$R_F = \frac{0.76}{9.11} = 0.0834 \Omega$$

$$R_r = \frac{5.82}{1.2} = 4.85 \Omega$$

CAUTIONS:

- All connections should be tight and right.
- Make circuit carefully as per diagram.
- Take reading carefully.

Questions:

1 What is Zener diode.

Ans: Zener diode are semiconductor devices that allows c/n to flow in both direction but specialize in current flowing in reverse direction

Q.2 How do you solve a Zener diode question?

ANS: The Zener will conduct maximum current when voltage is maximum it will conduct minimum when voltage is minimum

Q.3 What is the work of a Zener diode?

ANS: It is used for voltage regulatory purpose

Q.4 What is a Zener diode used as?

ANS: It is used as voltage regulator for regulation voltage across small ckt. shunt.

Q.5 How do you identify a Zener diode?

ANS: Zener diode have plastic coating along with strips which is of same colour coding as that of diode.

Marks Obtained	25
Faculty Signature	Mand